

No. of Functional Features Included in the Solution

Date	11 July 2026
Team ID	
Project Name	Rising waters - Flood Prediction using Machine Learning
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Form	Allows users to enter weather and rainfall details	Frontend(HTML/CSS)	Done	High
2	Data Validation	Validates input values before prediction	Flask Backend	Done	Medium
3	Flood Prediction	Predicts flood risk using trained ML model	ML Model	Done	High
4	Model Integration	Connects trained model with web application	Flask Backend	Done	High
5	Prediction Result Display	Displays prediction result to the user	Frontend	Done	Medium
6	Dataset Preprocessing	Cleans and prepares dataset for model training	Python	Done	Medium
7	Model Training	Trains Random Forest machine learning model	Scikit-learn	Done	High
8	Web Deployment	Deploys application on Render cloud platform	Deployment	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5

Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	User Input Form, Prediction Result Display
2	Backend / Logic	3	Data Validation, Model Integration, Prediction Processing
3	Database / Storage	1	Dataset Storage & Model File
4	API / Integration	1	Flask Prediction API
5	Security / Authentication	1	Input Validation & Error Handling